

# MUHAMMAD SAAD BIN MAZHAR

AI/ML Engineer | Generative AI | RAG & Agentic Systems | Full-Stack AI

Rawalpindi, Pakistan | +92-309-8318300 | msaadbinmazhar@gmail.com  
github.com/M-SAAD-BIN-MAZHAR | linkedin.com/in/m-saad-bin-mazhar | huggingface.co/Saadi55 | saad-s-portfolio-kappa.vercel.app

## PROFESSIONAL SUMMARY

MIT Hackathon Winner and AI/ML Engineer with 1.5+ years of production experience building and deploying enterprise-grade AI systems. 3rd-year Software Engineering student at NUST (CGPA 3.59/4.00). Specialized in Generative AI, RAG pipelines, agentic and multi-agent workflows, NLP, and computer vision, with full-stack delivery across healthcare, education, and SaaS. Published 2 fine-tuned HuggingFace models and a production npm MCP package. Strong in LangChain, LangGraph, CrewAI, FastAPI, Node.js, Docker, and AWS deployment.

## TECHNICAL SKILLS

**AI/ML:** Generative AI, LLMs, RAG, Fine-Tuning (LoRA/QLoRA), Prompt Engineering, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, CrewAI, AutoGen, Computer Vision, NLP, Transformers, BERT, OpenAI/Claude API, Embeddings, Vector Databases

**Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, NLTK, spaCy, Hugging Face Transformers

**Languages & Backend:** Python, JavaScript, C++, Java, SQL, FastAPI, Django, Flask, Node.js, Express.js, NestJS, REST API, WebSockets, SSE, JWT

**Data & Databases:** PostgreSQL, MySQL, MongoDB, Pinecone, FAISS, Pandas, NumPy, Feature Engineering

**MLOps & Cloud:** Docker, Kubernetes, CI/CD, MLflow, AWS (EC2, S3, SageMaker), Railway, Render, Vercel, n8n

**Tools:** Git, GitHub, Jupyter, Google Colab, Model Context Protocol (MCP), Streamlit, Linux

## PROFESSIONAL EXPERIENCE

### Freelance AI/ML Engineer - Independent Clients

Remote | Jan 2024 - Present

- Built and deployed 10+ production RAG pipelines and agentic workflows (LangChain, LangGraph, CrewAI), improving response accuracy by 87% via vector optimization and multi-stage retrieval.
- Published 2 fine-tuned NLP models on HuggingFace for text classification and sentiment analysis, achieving 18% accuracy gain over baseline using LoRA.
- Authored and published github-repo-manager-mcp, a production npm MCP package enabling AI assistants to manage GitHub repositories (12+ operations, 100% test coverage).
- Engineered real-time voice agents with sub-1.5s latency using asyncio, FAISS memory, multi-provider LLMs (OpenAI, Mistral, HuggingFace), and Silero VAD; reached 99.2% uptime.
- Deployed FastAPI and Node.js services on AWS, Railway, and Vercel with JWT auth, Docker, and CI/CD, cutting deployment time by 73%.

### AI/ML Engineering Intern - Elevvo Pathways

Remote | Aug 2025 - Sep 2025

- Built classification, clustering, and predictive models (scikit-learn, XGBoost) for business intelligence use cases.
- Improved CNN traffic-sign recognition accuracy by 8.5 points through data augmentation and architecture tuning.

### AI/ML Engineering Intern - Innoverse.pk

Remote | Aug 2025 - Sep 2025

- Built a depression-prediction system (Random Forest, Gradient Boosting) reaching 89.3% accuracy, deployed on Streamlit Cloud with a full preprocessing and feature-engineering pipeline.

## KEY PROJECTS

### Web Agent API - MIT Hackathon Winner (2026)

Engineered a browser-level AI SDK with real MCP integration and a 6-agent orchestration system (planning, navigation, extraction, execution, memory, voice). Built a Chrome Extension for DOM automation; deployed with sub-500ms response latency.

### Academix.io - AI Academic Automation Platform

Next.js, FastAPI, CrewAI, PostgreSQL, Docker, Railway

Built a full-stack multi-agent platform that auto-generates lab reports, runs MATLAB code, and exports DOCX. Reduced report creation from 8-10 hours to 4 minutes; supports 2000+ monthly active users.

### Job Radar - Intelligent Job Matching & Automation

n8n, OpenAI, Hunter.io, Apollo.io, HubSpot, FastAPI, React, PostgreSQL

Orchestrated 7 n8n workflows processing 96+ candidates daily at 99.5% reliability. Added GPT-4 scoring (-82% manual screening), smart enrichment routing (-67% API cost), and quote-aware WhatsApp command routing with 100% accuracy.

### BS Management System - Multi-Tenant SaaS

*FastAPI, PostgreSQL, SQLAlchemy, React, Docker, AWS, SSE*

Architected a multi-tenant business platform with RBAC, real-time SSE inventory sync (<100ms), and an agentic chatbot that cut manual data entry by 85%. Processes 1000+ daily transactions at 99.8% accuracy; 99.7% uptime.

### Intelligent Healthcare Platform - Full-Stack AI Medical System

*React, FastAPI, TensorFlow, XGBoost, RAG, PostgreSQL, AWS*

Built CNN pneumonia detection (94.2% accuracy), XGBoost risk assessment (89.5% AUC), and a RAG medical chatbot with citation-based confidence. Deployed on AWS.

## OPEN SOURCE & PUBLISHED WORK

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- HuggingFace (Saadi55): 2 fine-tuned NLP models, +18% accuracy with LoRA, -55% model size.
- github-repo-manager-mcp (npm): Production MCP server, 12+ GitHub operations, 100% test coverage, Claude Desktop compatible.
- GitHub: 91+ repositories spanning AI/ML, full-stack, and automation.

## EDUCATION

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### B.E. Software Engineering

National University of Sciences and Technology (NUST), Islamabad | 2024 - 2028

CGPA: 3.59/4.00 | Dean's List

*Coursework: Machine Learning, Data Structures, Algorithms, Database Systems, Computer Networks, Operating Systems*

## ACHIEVEMENTS

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- MIT Hackathon Winner (2026) - Web Agent API with real MCP integration and 6-agent system.
- Founder of a production multi-tenant SaaS platform onboarding enterprise clients.
- Delivered production AI systems across healthcare, education, and business domains.
- Published 2 HuggingFace models, 1 npm package, and 91+ GitHub repositories.

## CERTIFICATIONS

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Advanced RAG with Vector Databases (IBM/Coursera); ChatGPT Prompt Engineering for Developers (DeepLearning.AI); Introduction to Generative AI (AWS); Data Science & AI Advanced (Udemy).